

9. Quantum Chromodynamics

Revised August 2023 by J. Huston (Michigan State U.), K. Rabbertz (KIT) and G. Zanderighi (MPP, Garching; München Tech. U.).

9.1	Basics	1
9.1.1	Running coupling	2
9.1.2	Quark masses	3
9.2	Structure of QCD predictions	4
9.2.1	Fully inclusive cross sections	4
9.2.2	Processes with initial-state hadrons	5
9.2.3	Cross sections with phase-space restrictions	10
9.2.4	Accuracy of predictions	17
9.3	Experimental studies of QCD	19
9.3.1	Hadronic final-state observables	20
9.3.2	QCD measurements at colliders	23
9.4	Determinations of the strong coupling constant	30
9.4.1	Hadronic τ decays and low Q^2 continuum:	31
9.4.2	Heavy quarkonia decays:	32
9.4.3	PDF fits:	32
9.4.4	Hadronic final states of e^+e^- annihilations:	34
9.4.5	Observables from hadron-induced collisions:	35
9.4.6	Electroweak precision fit:	37
9.4.7	Lattice QCD:	38
9.4.8	Determination of the world average value of $\alpha_s(m_Z^2)$:	39

9.1 Basics

Quantum Chromodynamics (QCD), the gauge field theory that describes the strong interactions of colored quarks and gluons, is the SU(3) component of the SU(3)×SU(2)×U(1) Standard Model of Particle Physics. The Lagrangian of QCD is given by

$$\mathcal{L} = \sum_q \bar{\psi}_{q,a} (i\gamma^\mu \partial_\mu \delta_{ab} - g_s \gamma^\mu t_{ab}^C \mathcal{A}_\mu^C - m_q \delta_{ab}) \psi_{q,b} - \frac{1}{4} F_{\mu\nu}^A F^{A\mu\nu}, \quad (9.1)$$

where repeated indices are summed over. The γ^μ are the Dirac γ -matrices. The $\psi_{q,a}$ are quark-field spinors for a quark of flavor q and mass m_q , with a color-index a that runs from $a = 1$ to $N_c = 3$, *i.e.* quarks come in three “colors.” Quarks are said to be in the fundamental representation of the SU(3) color group.

The $\mathcal{A}_\mu^C$ correspond to the gluon fields, with C running from 1 to $N_c^2 - 1 = 8$, *i.e.* there are eight kinds of gluon. Gluons transform under the adjoint representation of the SU(3) color group. The t_{ab}^C correspond to eight 3×3 matrices and are the generators of the SU(3) group (*cf.* the section on “SU(3) isoscalar factors and representation matrices” in this *Review*, with $t_{ab}^C \equiv \lambda_{ab}^C/2$). They encode the fact that a gluon’s interaction with a quark rotates the quark’s color in SU(3) space. The quantity g_s (or $\alpha_s = \frac{g_s^2}{4\pi}$) is the QCD coupling constant. Besides quark masses, which have

electroweak origin, it is the only fundamental parameter of QCD. Finally, the field tensor $F_{\mu\nu}^A$ is given by

$$\begin{aligned} F_{\mu\nu}^A &= \partial_\mu \mathcal{A}_\nu^A - \partial_\nu \mathcal{A}_\mu^A - g_s f_{ABC} \mathcal{A}_\mu^B \mathcal{A}_\nu^C, \\ [t^A, t^B] &= if_{ABC} t^C, \end{aligned} \quad (9.2)$$

where the f_{ABC} are the structure constants of the SU(3) group.

Neither quarks nor gluons are observed as free particles. Hadrons are color-singlet (*i.e.* color-neutral) combinations of quarks, anti-quarks, and gluons.

Ab-initio predictive methods for QCD include lattice gauge theory and perturbative expansions in the coupling. The Feynman rules of QCD involve a quark-antiquark-gluon ($q\bar{q}g$) vertex, a 3-gluon vertex (both proportional to g_s), and a 4-gluon vertex (proportional to g_s^2). A full set of Feynman rules is to be found for example in Refs. [1, 2].

Adopting a standard notation where repeated indices are summed over, useful color-algebra relations include: $t_{ab}^A t_{bc}^A = C_F \delta_{ac}$, where $C_F \equiv (N_c^2 - 1)/(2N_c) = 4/3$ is the color factor (“Casimir”) associated with gluon emission from a quark; $f_{ACD} f_{BCD} = C_A \delta_{AB}$, where $C_A \equiv N_c = 3$ is the color factor associated with gluon emission from a gluon; $t_{ab}^A t_{ab}^B = T_R \delta_{AB}$, where $T_R = 1/2$ is the color factor for a gluon to split to a $q\bar{q}$ pair.

There is freedom for an additional CP -violating term to be present in the QCD Lagrangian, $\theta \frac{\alpha_s}{8\pi} F^{A\mu\nu} \tilde{F}_{\mu\nu}^A$, where θ is an additional free parameter, and $\tilde{F}_{\mu\nu}^A$ is the dual of the gluon field tensor, $\frac{1}{2} \epsilon_{\mu\nu\sigma\rho} F^{A\sigma\rho}$, with $\epsilon_{\mu\nu\sigma\rho}$ being the fully antisymmetric Levi-Civita symbol. Experimental limits on the electric dipole moment (EDM) of ultracold neutrons [3, 4] and atomic mercury [5] constrain the QCD vacuum angle to satisfy $|\theta| \lesssim 10^{-10}$. For an overview of current and future experiments measuring EDMs see [6]. Further discussion is to be found in Ref. [7] and in the Axions section in the Listings of this *Review*.

This section will concentrate mainly on perturbative aspects of QCD as they relate to collider physics. Related textbooks and lecture notes include Refs. [1, 2, 8–11]. Aspects specific to Monte Carlo event generators are reviewed in the dedicated section 43. Lattice QCD is also reviewed in a section of its own, Sec. 17, with further discussion of perturbative and non-perturbative aspects to be found in the sections on “Quark Masses”, “The CKM quark-mixing matrix”, “Structure functions”, “Fragmentation Functions”, “High Energy Soft QCD and Diffraction”, “Passage of Particles Through Matter” and “Heavy-Quark and Soft-Collinear Effective Theory” in this *Review*.

9.1.1 Running coupling

In the framework of perturbative QCD (pQCD), predictions for observables are expressed in terms of the renormalized coupling $\alpha_s(\mu_R^2)$, a function of an (unphysical) renormalization scale μ_R . When one takes μ_R close to the scale of the momentum transfer Q in a given process, then $\alpha_s(\mu_R^2 \simeq Q^2)$ is indicative of the effective strength of the strong interaction in that process.

The coupling satisfies the following renormalization group equation (RGE):

$$\mu_R^2 \frac{d\alpha_s}{d\mu_R^2} = \beta(\alpha_s) = -(b_0 \alpha_s^2 + b_1 \alpha_s^3 + b_2 \alpha_s^4 + \dots), \quad (9.3)$$

where $b_0 = (11C_A - 4n_f T_R)/(12\pi) = (33 - 2n_f)/(12\pi)$ is referred to as the 1-loop β -function coefficient, the 2-loop coefficient is $b_1 = (17C_A^2 - n_f T_R(10C_A + 6C_F))/(24\pi^2) = (153 - 19n_f)/(24\pi^2)$, and the 3-loop coefficient is $b_2 = (2857 - \frac{5033}{9}n_f + \frac{325}{27}n_f^2)/(128\pi^3)$ for the SU(3) values of C_A and C_F . Here n_f is the number of quark flavors. The 4-loop coefficient, b_3 , is to be found in Refs. [12, 13], while the 5-loop coefficient, b_4 , is in Refs. [14–18]. The coefficients b_2 and b_3 (and beyond) are renormalization-scheme-dependent and given here in the modified minimal subtraction scheme ($\overline{\text{MS}}$) [19], by far the most widely used scheme in QCD and the one adopted in the following.